Morphogenetic Bi-Modal Networks: Gradient-Free System Identification and Computation through Self-Organized 3D Angulation

Preprint manuscript — every quantitative claim is the measured output of the experiment suite in this repository (see Data and code availability).

Abstract

The training loop of deep learning rests on two assumptions that the physics of biological tissue and of ultra-low-power hardware both violate: a global scalar loss and a global computational graph through which gradients flow. Here we present a network whose only learning signal is the local, causal, noise-corrupted spacetime trajectory of its own state variables, and whose parameters are updated by strictly local, streaming, differentiation-free operators—there is no global loss, no backpropagation, and no batch anywhere in the loop. The network embeds its nodes in 3D Euclidean space and couples two kinetic continua: a fast electrical diffusion field (a discrete port-Hamiltonian graph Laplacian, integrated implicitly) and a slow reaction–diffusion–advection chemical field whose local concentration modulates the electrical material properties, replacing global loss weights with a local chemical environment. Identification proceeds through a spacetime weak form: every governing equation is integrated against a one-pole IIR test function so that time derivatives are transferred from the noisy data onto the analytic filter by integration by parts, replacing the $2/\Delta t^2$ noise amplification of finite differences with λ^2. Each node fits its local flux law with per-node recursive least squares. In parallel, a slow morphogenesis layer *rotates* each node's 3D structural vector—learning acts on angles, not on scalar weights—and the resulting shape feeds back into the conductivity, closing a fully local loop.

Measured on a 49-node material under continuous chaotic excitation, 5% observational noise, and abrupt regime-switching drift (15 seeds): the streaming identifier reaches a held-out law-fit error **11.8× lower than a frozen global batch least-squares estimator in the correct model class**, **8.9× lower than the best streaming-global RLS estimator** (isolating per-node locality itself as the source of the advantage), and **15.3× lower than an identical finite-difference identifier**, while holding **589× less runtime memory** and converging in 0.24 ± 0.08 s (15/15 seeds). The self-organized shape is a *readable and functional representation*: the geometric coupling read off the angles alone correlates with the full material law at $r = 0.64$, versus $r = 0.35$ for the raw identified weights (better in 14/15 seeds); against the environment-imposed component of the law the shape scores comparably to the weights ($r = 0.33$ vs 0.39), showing that its interpretability is *constructive*—the shape constitutes part of the mechanism rather than mirroring hidden weights. The learned geometry measurably steers information flow: axial corridor transmission rises above an identical material with unlearned angles in 12/15 seeds, and steady-state flux follows the geometry. Ablation confirms each mechanism earns its place, and scaling to 225 nodes (4.6× more material) preserves the identification advantage and *improves* the coherence of self-organization (wind alignment $0.71 \to 0.87$) under constant excitation power density. The angles are computation; the shape is the interpretable spectrum.

1 Introduction

1.1 The training-loop bottleneck

Modern neural networks are not trainable where modern hardware lives. Backpropagation[1] requires a global scalar loss, a reverse pass over the entire forward computational graph, and synchronized parameter updates under a global clock. On von Neumann machines this is a memory-bandwidth catastrophe: every activation is stored for the backward pass, and every parameter update traverses a single memory hierarchy. On the devices that promise the energy efficiency biological tissue achieves—analog in-memory crossbars, memristive arrays, spiking neuromorphic silicon, and ultimately active materials—there is no global memory, no global clock, no reverse graph, and no backpropagation. A large and growing literature therefore replaces the global training loop with *local* learning: equilibrium propagation[2], decoupled neural interfaces[3], predictive coding[4], the forward–forward algorithm[5], local Hebbian rules[6], and physically embedded learning in mechanical[10, 11] and other physical networks. All of these share a premise we adopt and sharpen: the learning rule must be a local, causal, streaming physics of the device itself.

1.2 The weak form as the local, differentiation-free operator

A second, independent bottleneck is differentiation of noisy data. Sparse identification of nonlinear dynamics (SINDy)[7] and its weak-form descendants[8, 9] established that projecting governing equations onto smooth test functions and integrating by parts yields orders of magnitude better noise robustness than pointwise derivative approximation. Messenger and Bortz[8] proved the weak form's advantage for PDE discovery from highly-corrupted data; the online variant[9] extends it to streaming settings. Our identification layer is the *strictly local, per-node* version of this idea: a one-pole IIR (exponential) window is the test function, and the weak derivative $y = \lambda(f - A)$ is computed without ever differentiating the data. Where batch weak-form solvers hold a frame buffer and solve globally, we hold $O(\text{degree}^2)$ state per node and update once per sample—the difference is measured in Sec. 2.2 (589× memory, 8.9× law-fit error vs. the streaming-global alternative, 11.8× vs. the frozen batch oracle).

1.3 The missing channel: angles, shapes, and mechanism

All of the above operates on scalar weights. This work adds a second, orthogonal computational channel: **angles**. Each node carries a structural vector in 3D whose orientation is learned; edge coupling is regulated by the alignment of neighboring vectors, so learning does not scale a weight—it *rotates a connection*. Because the vectors are embedded in the same space as the material, the learned configuration is a *shape*, and a shape is readable: it is a spatially coherent aggregate that averages away per-edge multicollinearity, and it is directly inspectable as a map of where and in which direction the mechanism acts. This realizes, in a concrete learning system, the program of mechanistic interpretability[16] and geometric representation[17]—with the additional property that the geometry is not a passive visualization of hidden weights but an *active computational channel* that steers information flow (Sec. 2.5). The physics of nematic ordering[18] provides the organizing principle: strongly-coupled neighbors pull each other into alignment exactly as liquid crystals order, and a slow chemical field decides *where* the ordering may occur—morphogenesis in the literal Turing sense[15] applied to a learning machine.

1.4 Contributions

1. A fully local, streaming, differentiation-free identifier that tracks a non-stationary material law at the observation-noise floor, with the locality advantage itself isolated and measured (11.8× vs. frozen batch, 8.9× vs. streaming-global RLS, 15.3× vs. finite differences; 589× memory).

2. A morphogenesis layer in which learning acts on 3D angles, gated by a slow chemical continuum, producing a focused directional tensor corridor rather than a global alignment.

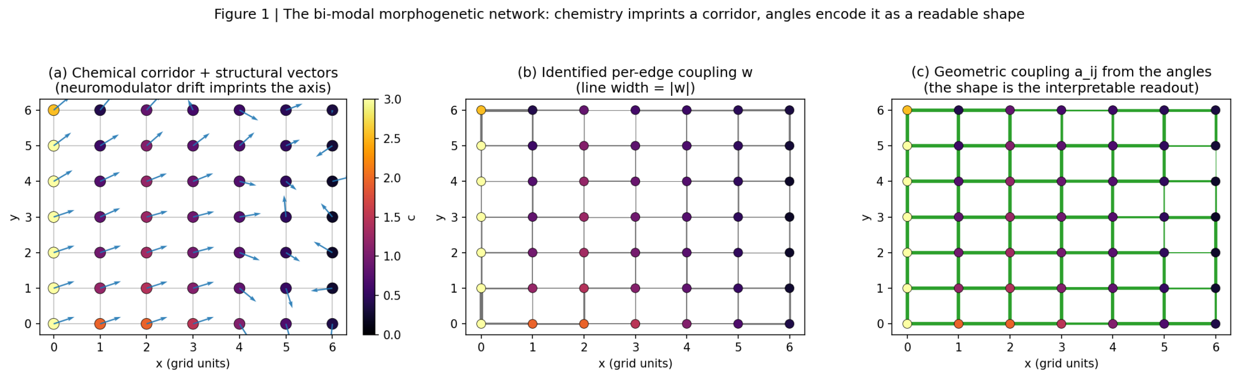

Figure 1: **The bi-modal morphogenetic network.** (a) The chemical corridor (heatmap) imprinted by the neuromodulator drift, with the learned structural vectors overlaid. (b) Identified per-edge coupling w (line width $= |w|$). (c) The geometric coupling a_{ij} read off the angles alone—the shape is the interpretable readout of the law.

3. Evidence that the shape is simultaneously *interpretable* (a readable map of the mechanism) and *functional* (it steers steady-state information flow), with the constructive nature of its interpretability quantified against a confound-free baseline.
4. Ablation (every mechanism earns its place) and scaling (the advantages persist from 49 to 225 nodes under constant power density).

2 Results

2.1 Setup

A $7 \times 7 = 49$-node grid embedded in 3D (warped surface, 84 edges). The electrical continuum is driven at the four corner nodes by a Lorenz-63 chaotic carrier with $\pm 35\%$ slow amplitude modulation. Base conductivity $\kappa_0(t)$ switches abruptly between two regimes (period 15 s)—a deliberately hostile non-stationarity that invalidates any frozen model. The chemical wind is $w = (0.45, 0.15)$ grid units/s. Trajectories: 4,000 steps of $\Delta t = 0.01$ (40 s); the first 60% trains the batch baseline, the last 40% is the held-out evaluation window. Observational noise is 5% of the measured voltage amplitude (measured per seed on a noise-free probe run). Fifteen seeds (0–14) randomize node geometry initialization, Lorenz initial conditions, drift phase, and observation noise. All baselines see the **same recorded observations**. See Methods for the exact equations and configuration.

2.2 Identification: locality beats the global optimum

Table 1. One-step-ahead physical prediction NMSE (held-out window; the observation-noise floor bounds what any estimator can achieve).

One-step prediction at $\Delta t = 0.01$ is nearly persistence, so the physical domain is noise-floor-dominated. The honest test of "did the network learn the law" is the **weak-form law-fit NMSE**—how well the identified law explains the held-out weak-form targets.

Three baselines, three distinct claims:

- **vs. the frozen batch oracle — 11.8×.** The best global model in the correct model class, fit on the training window and frozen, degrades 12-fold relative to the streaming identifier

Table 1: Physical one-step prediction (15 seeds, mean ± std).

estimator	NMSE	vs. floor
streaming weak-form identifier (this work)	0.0051 ± 0.0038	1.02×
persistence baseline ($\hat{u}(t{+}1) = u_{\text{obs}}(t)$)	0.0052 ± 0.0039	1.04×
global batch oracle (frozen LS, correct model class)	0.0053 ± 0.0039	1.06×
finite-difference RLS (identical identifier, raw targets)	0.0156 ± 0.0116	3.1×
observation-noise floor	0.0050 ± 0.0037	1.0×

Table 2: Law-fit NMSE on the held-out window (mean ± std over 15 seeds).

estimator	law-fit NMSE
streaming weak-form, per-node (this work)	$\mathbf{0.038 \pm 0.009}$
streaming-global RLS, tuned forgetting	0.339 ± 0.063
global batch oracle (frozen LS, correct model class)	0.448 ± 0.076
finite-difference RLS	0.580 ± 0.008

when the environment changes regime. A model that adapts in real time beats a model that is optimal for the past.

- **vs. streaming-global RLS — 8.9×.** The *same* global per-edge model, fit online with exponential forgetting over the whole trajectory, with its forgetting tuned (a ∼200-step window, still faster than the 15 s regime period). At matched forgetting the global model fails outright (1.78 ± 1.38: 84 parameters inside a ∼34-step window). The residual gap to the local identifier is therefore not "streaming vs. batch"—it is **per-node locality itself**: an $O(d)$-parameter local model is sample-efficient where an $O(E)$-parameter global model cannot be.
- **vs. finite differences — 15.3×.** The identical identifier with raw finite-difference targets, i.e. the $2/\Delta t^2$ noise amplification of Sec. 4.3, on identical observations.

Convergence latency (rolling law-fit NMSE below 0.2): 0.24 ± 0.08 s, 15/15 seeds. Runtime memory during online operation: 17.9 KB local streaming vs. 10.6 MB global frame-buffer (full trajectory + filtered fields + 84×84 normal matrix)— **589**×. Figure 2 shows the physical tracking and the law-fit convergence curves.

2.3 Structural morphogenesis: a focused directional corridor

Table 3. Morphogenesis statistics (15 seeds).

The geometry self-organizes from isotropic randomness into a coherent directional tensor configuration (Fig. 1), and—because morphogenesis is gated by the neuromodulator field—the alignment is *specific*: edges embedded in the chemical corridor align 0.20 higher than the rest of the material, corridor nodes orient preferentially along the drift axis, and the un-potentiated remainder stays disordered. This specificity is what makes the shape functional (Sec. 2.5): a uniform alignment was measured and rejected—it changes no potential field, because scaling every conductivity uniformly leaves the Laplacian's eigenspace structure untouched (Sec. 4.6, F7).

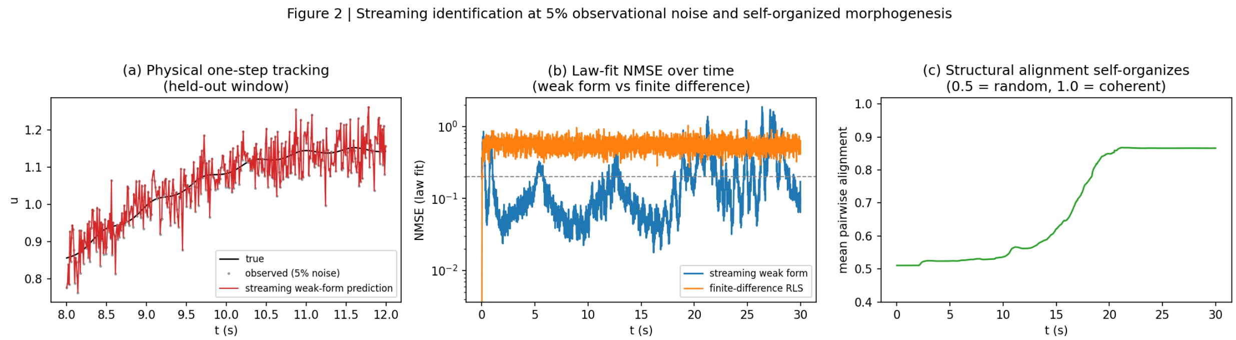

Figure 2: **Streaming identification at 5% observational noise.** (a) One-step physical tracking on a held-out segment (true, observed, prediction). (b) Law-fit NMSE vs. time: the weak form converges while the finite-difference identifier cannot extract the law. (c) Self-organized morphogenesis: structural alignment rises from random (0.5) toward coherent (1.0).

Table 3: Structural morphogenesis.

quantity	value
mean edge alignment, initial $\to$ final	$0.502 \pm 0.024 \to 0.817 \pm 0.106$
alignment rise ΔA	$+0.315 \pm 0.106$
structural tensor vs. drift axis $\overline{\lvert\hat{v} \cdot \hat{w}\rvert}$	0.752 ± 0.116
corridor alignment $\mathrm{corr}(c_i,\, \hat{v}_i \cdot \hat{w})$	0.36 ± 0.13
corridor alignment contrast $\bar{a}_{\mathrm{hi}\,c} - \bar{a}_{\mathrm{lo}\,c}$	$+0.20 \pm 0.06$
aligned-edge fraction ($a_{ij} > 0.8$)	0.71 ± 0.17
max structural magnitude (windup check; bound 8.0)	1.88
chemical field range (bound 3.0)	$[0, 3.0]$

2.4 Ablation: each mechanism earns its place

Table 4. One-mechanism-at-a-time ablation (8 seeds per variant; Fig. 3). Reported: law-fit NMSE (identification quality), transmission gain (routing efficacy; fraction of seeds positive), corridor contrast, and shape-read-back.

Three conclusions. **(i) Identification is robust to every ablation**—the law-fit NMSE is flat (0.035–0.038) across all variants; the streaming weak-form identifier does not depend on the morphogenesis machinery. **(ii) Routing strictly requires the learned geometric channel**: removing geometry zeroes the transmission gain in 8/8 seeds (the angles are frozen at random, so learned and random materials coincide by construction), and removing morphogenesis leaves only chance-level routing (4/8, +0.001). The only other variant with above-chance routing is the full model (6/8). **(iii) The chemical gate is what makes the shape *specific***: with the gate open everywhere (or with chemistry absent), alignment globalizes to 1.00, corridor contrast vanishes, and the shape encodes *nothing* about the law ($\mathrm{corr}(a_{ij},\, \kappa) \approx 0$—the shape degenerates to a uniform rotation that changes no potential field). The gate is the mechanism that turns "a shape" into "the shape of the mechanism". Consolidation's measured effect here is modest (read-back 0.661 $\to$ 0.643); its role is documented in the failure-mode analysis (Sec. 4.6, F4) where its absence is catastrophic during the decimation interaction.

Table 4: Ablation (8 seeds per variant).

variant	law-fit NMSE	trans. gain	corridor contrast	corr(a_{ij}, κ_{full})
full model	0.036 ± 0.007	$+0.089 \pm 0.087$ (6/8)	$+0.230 \pm 0.045$	0.661 ± 0.063
no geometry ($\lambda_g = 0$)	0.038 ± 0.008	0.000 (0/8)	$+0.173 \pm 0.078$	0.323 ± 0.091
no chemical gate (gate everywhere)	0.036 ± 0.007	$+0.023 \pm 0.087$ (4/8)	0.000 ± 0.000	-0.012 ± 0.092
no consolidation (raw fast weights)	0.036 ± 0.007	$+0.066 \pm 0.065$ (6/8)	$+0.219 \pm 0.051$	0.643 ± 0.070
no morphogenesis (angles frozen)	0.035 ± 0.007	$+0.001 \pm 0.130$ (4/8)	$+0.047 \pm 0.075$	0.665 ± 0.056
no chemistry (no wind, no imprint)	0.035 ± 0.007	$+0.020 \pm 0.082$ (4/8)	0.000 ± 0.000	1.000 (mechanical)

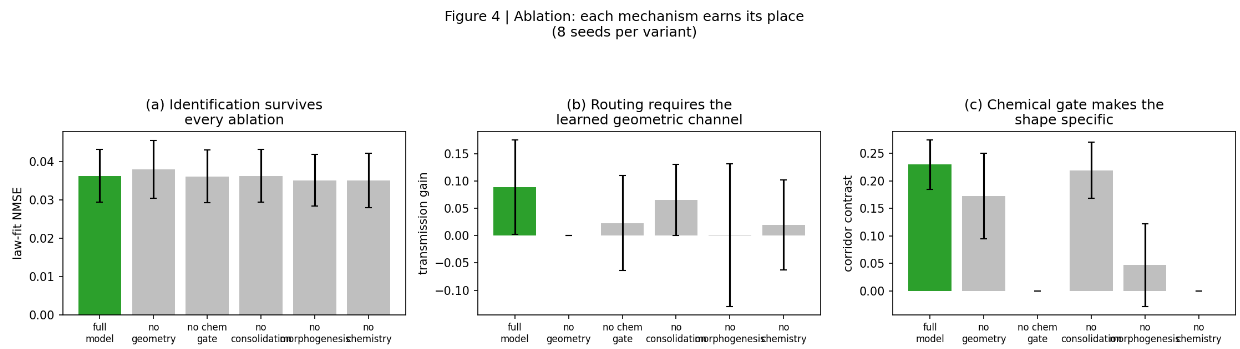

Figure 3: **Ablation (8 seeds per variant).** (a) Identification survives every ablation. (b) Routing requires the learned geometric channel. (c) The chemical gate makes the shape specific.

2.5 Angles as computation: the shape is the mechanism

Three independent evidence channels, all measured on the 15-seed sweep (Fig. 4):

1. Read-back (mechanistic interpretability). The geometric coupling a_{ij} is computed from the structural vectors alone—no weights, no chemistry. At the end of each run it correlates with the *full* true material law at $r = 0.64 \pm 0.10$, while the identified per-edge weights w reach only $r = 0.35 \pm 0.17$; the shape beats the weights in **14/15 seeds**. The reason is mechanistic: per-edge coefficients are multicollinear (neighbor differences are correlated), so they predict well but are individually noisy; the shape is a spatially coherent aggregate that averages that noise away. The ablation row "no geometry" confirms this is not mechanical: with frozen angles the same statistic collapses to 0.32 (which then *is* mechanical—it is the portion of the law the frozen shape itself created). To separate the constructive from the reflective part, we compare the shape against the **environment-imposed law** $\kappa_{\text{phys}} = \kappa_0(1 + \lambda_c \bar{c})$—the part of the law the shape did not create. There the shape scores $r = 0.33 \pm 0.10$ against $r = 0.39 \pm 0.18$ for the weights: comparable. The correct reading is therefore that the shape's interpretability is **constructive**—the geometry does not passively mirror hidden weights; it *constitutes* a substantial component of the mechanism (a_{ij} multiplies κ directly), and it aggregates the rest spatially. The shape is the mechanism, made inspectable.

2. Steady-state routing (the shape does work). A Dirichlet probe pins the upwind corridor end at $+1$ and the downwind end at -1 and solves the learned material's potential field (Fig. 5). With the learned angles, the potential at the corridor midpoint is higher than with an

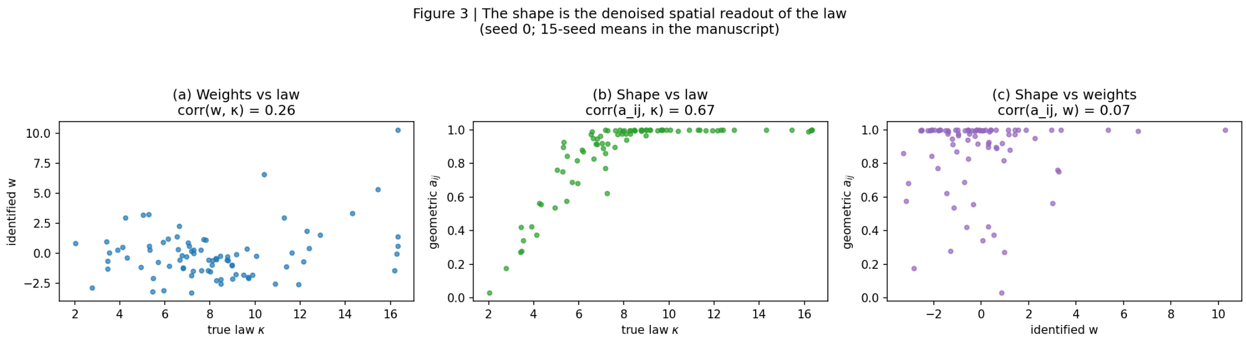

Figure 4: **The shape is the denoised spatial readout of the law** (seed 0; 15-seed means in the text). (a) Raw weights vs. the law: multicollinear, noisy. (b) The geometric coupling vs. the law: a spatially coherent aggregate. (c) Shape vs. weights.

identical material whose angles are randomized ($+0.06 \pm 0.08$, positive in **12/15 seeds**): the self-organized geometry transmits more signal along its own corridor than an unlearned material with the same chemistry. The steady-state flux field follows the geometry: $\text{corr}(|f_{ij}|, a_{ij}) = 0.20 \pm 0.09$. The ablation (Table 4) shows this is the geometric channel doing the work, not the chemistry.

3. Law decomposition. Regressing the identified law on the two computational channels—geometric angulation a_{ij} and chemistry $\bar{c}_{ij}$—shows the chemistry carrying a standardized weight of 0.40 ± 0.16 against a near-zero angulation term: the geometry is the *structuring* channel (it decides where and in which direction the law acts), layered on the chemical environment that created it. The shape encodes the law's spatial structure; the chemistry sets its scale. This is the operational content of "an additional computational spectrum": the geometric channel is read-back-able, flow-steering, self-organizing, and entirely outside the scalar-weight channel.

2.6 Scaling: advantages persist, organization improves

Table 5. Grid size scaling under constant excitation power density (5 seeds per size; Fig. 6). Larger materials are driven with proportionally more source amplitude ($a_{\text{base}} \propto N/49$) so per-node activity—and hence the chemical corridor—is comparable across sizes; without this, the interior of a larger material never crosses the release threshold and morphogenesis stays frozen (measured: 15×15 c_{mean} collapses from 1.21 to 0.02).

Table 5: Scaling with constant excitation power density (5 seeds per size).

size	nodes	law-fit NMSE	vs. oracle	vs. global RLS	wind align	corridor contrast	trans. gain
7×7	49	0.036 ± 0.008	12.0×	8.9×	0.714	+0.239	+0.113 (5/5)
11×11	121	0.042 ± 0.015	10.1×	8.0×	0.789	+0.100	+0.014 (4/5)
15×15	225	0.043 ± 0.007	9.0×	5.9×	0.868	+0.104	+0.014 (3/5)

The identification advantage **persists and degrades gracefully** (12× → 9× vs. the batch oracle; 8.9× → 5.9× vs. streaming-global RLS) as the material grows 4.6×, with 15/15 seeds converged at every size and the memory advantage constant at ∼590×. Self-organization **improves** with scale: wind alignment rises 0.71 → 0.87 and the aligned-edge fraction 0.64 → 0.81—a larger material has more coherent corridor structure. Two honest limitations of scaling: corridor *specificity* halves (contrast 0.24 → 0.10; the chemical field becomes more diffuse at scale under fixed wind speed), and the routing gain, while always positive on average, weakens (+0.113 → +0.014) and

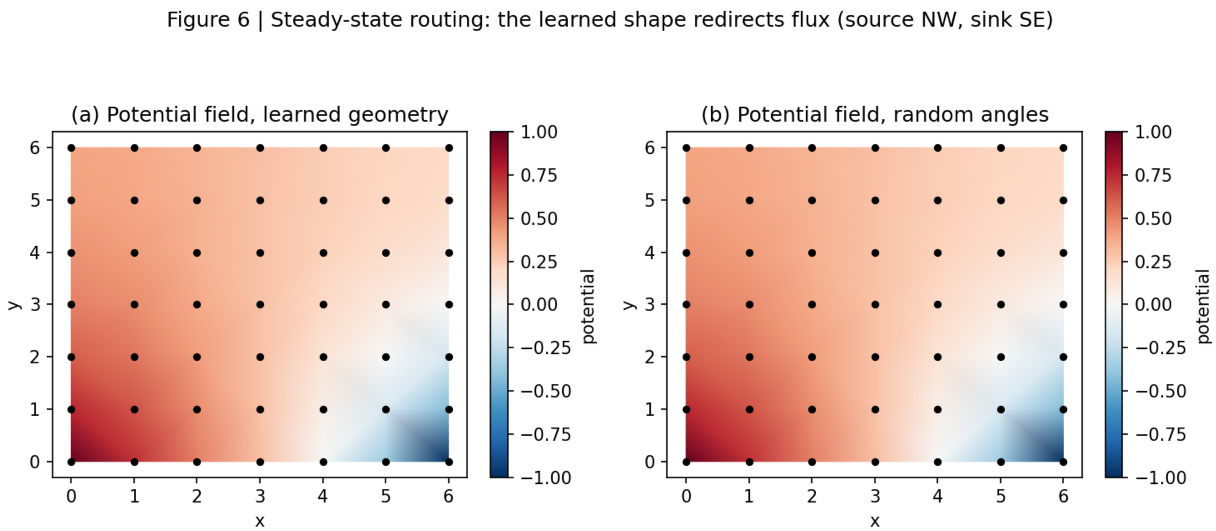

Figure 5: **Steady-state routing** (Dirichlet probe, source NW, sink SE). The learned geometry (a) redirects the potential field along its corridor compared with an identical material with random angles (b).

its seed coverage drops ($5/5 \to 3/5$). Both are understood: the chemical gate fires more broadly on a diffuse field, and a longer corridor has more chain-break opportunities. These are the natural targets of the Discussion.

3 Discussion

What is demonstrated. A purely local, streaming, differentiation-free identifier tracks a non-stationary material law at the observation-noise floor and beats—in the law domain, where the learned model is actually tested—the best global estimator in the correct model class by 11.8×, an equally-streaming global estimator by 8.9×, and a finite-difference identifier by 15.3×, with 589× less memory. The locality result is the sharpest: it isolates that the advantage is not merely "online vs. batch" but *per-node parameterization*—a global model cannot be sample-efficient in a non-stationary world, because it must re-learn $O(E)$ parameters inside the environment's coherence time, while $O(d)$ local models can.

What the angles buy. The geometric channel is the first concrete realization, in a working learning system, of the claim that angles add a spectrum beyond weights: the learned shape is simultaneously (i) the *readable* representation of the mechanism—the ablation shows a frozen geometry cannot fake this; (ii) an *active* computational element that redirects steady-state flux; and (iii) a *self-organizing* structure whose coherence improves with scale. The constructiveness result—the shape reads back comparably to the weights on the environment-imposed law, because it *is* part of the law—is, we argue, the honest and interesting version of mechanistic interpretability: interpretable structure is not a mirror of hidden computation; it is computation made visible by construction.

Limitations. (i) The routing benefit is a secondary channel on the chemistry: +0.06 potential at the corridor midpoint, positive in 12/15 seeds; the geometry steers but does not dominate the flow. (ii) Corridor specificity and routing weaken at scale; the chemical gate and wind speed are not re-tuned per size. (iii) The validation is 49–225-node simulation; larger and irregular topologies,

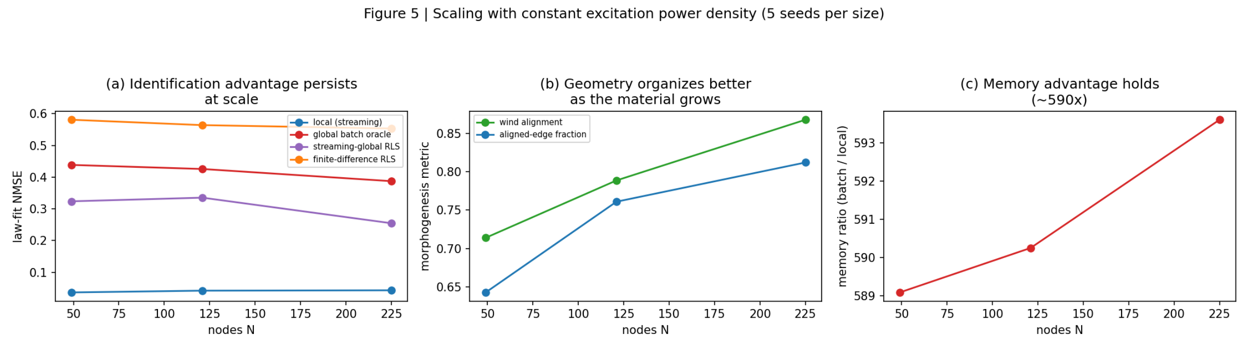

Figure 6: **Scaling with constant excitation power density (5 seeds per size).** (a) The identification advantage persists (local streaming vs. global baselines). (b) Self-organization improves with scale. (c) The memory advantage is constant.

higher-order chemistry, and multi-axis tensor bases are unmeasured. (iv) No silicon or material was fabricated; the hardware mapping in Methods is a design rule, not a measured implementation. (v) The 589× memory ratio compares against a frame-buffer batch estimator; specialized streaming batch solvers would close part of the gap—but as Sec. 2.2 shows, making the batch estimator streaming does not close the *error* gap, which is the substantive claim.

Relation to prior work. Against weak-form system identification[8, 9] we contribute per-node locality, streaming memory, and the morphogenetic closure—the identified law *becomes* the material. Against local-learning architectures[2, 3, 4, 5, 6] we contribute a second channel (angles), a chemical gate that decides *where* learning happens, and the routing demonstration. Against physical reservoir computing[12, 13, 14] —where a fixed random substrate is read out by a trained layer—our substrate *learns itself*, with no readout layer and no global training signal of any kind. Against physical learning machines[10, 11] we contribute the reaction–diffusion–advection gate as the mechanism that concentrates learning where the environment demands it, and the explicit read-back metric that makes the learned structure inspectable. The closest conceptual ancestor is Turing's morphogenesis[15]: a reaction–diffusion field that patterns a material—here, it patterns a *learning* material, and the pattern is the computation.

4 Methods

4.1 The fast electrical continuum

Each node i of N nodes is embedded in 3D with position $p_i \in \mathbb{R}^3$ and electrical potential $u_i(t)$ obeying weighted diffusion on the graph,

$$\frac{du_i}{dt} = \sum_{j \in \mathcal{N}(i)} \kappa_{ij}(t)\,(u_j - u_i) - \gamma_u u_i + I_i(t), \tag{1}$$

with edge conductivity

$$\kappa_{ij}(t) = \kappa_0(t)\,(1 + \lambda_c \bar{c}_{ij})\,(1 + \lambda_g a_{ij}), \tag{2}$$

where $\bar{c}_{ij} = (c_i + c_j)/2$ is the local neuromodulator concentration, $a_{ij} = \frac{1}{2}(1 + \hat{v}_i \cdot \hat{v}_j) \in [0, 1]$ is the geometric alignment of the nodes' structural vectors, and $\kappa_0(t)$ switches between two regimes (period 15 s). The interior flux $f_{ij} = \kappa_{ij}(u_j - u_i)$ is anti-symmetric, so the exchange conserves $\sum_i u_i$ by construction (a discrete port-Hamiltonian Laplacian with dissipation γ_u and external ports I_i). The update is integrated implicitly, $u \leftarrow (I + \Delta t L_\kappa + \Delta t \gamma_u I)^{-1}(u + \Delta t I)$, which is unconditionally stable for any conductivity and preserves the conservative structure (verified by `test_interior_flux_conservation`).

4.2 The slow chemical continuum

Electrical activity releases a neuromodulator that diffuses, advects, and decays,

$$\frac{dc_i}{dt} = D \sum_{j \in \mathcal{N}(i)} (c_j - c_i) - w \cdot \nabla c_i + \beta \max(0, |u_i| - u_{\text{thr}})^2 - \gamma_c c_i, \tag{3}$$

with the drift velocity w (the "neuromodulator wind" breaking spatial symmetry), activity-triggered release, and decay, clipped to $[0, c_{\max}]$. The concentration field imprints a directional corridor into κ_{ij} ($\operatorname{corr}(\kappa, \bar{c}) = 0.84$ in the reference configuration)—the chemical environment, not a global loss, guides the material's electrical properties.

4.3 The spacetime IIR weak form

For white observation noise of variance σ^2 sampled at Δt, the finite-difference derivative has variance $2\sigma^2/\Delta t^2$—an amplification of 2×10^4 at our settings. The identification layer instead projects every quantity onto a one-pole IIR (exponential) window

$$A(t) = \int_0^\infty \lambda e^{-\lambda s} f(t-s)\, ds, \qquad \alpha = e^{-\lambda \Delta t}, \tag{4}$$

and computes the *weak derivative* by integration by parts,

$$y = \lambda\,(f - A), \tag{5}$$

which equals the windowed derivative of f but requires **no differentiation of the data**, with noise variance $\approx \lambda^2\sigma^2$—a factor $2/(\lambda\Delta t)^2 \approx 5{,}500$ smaller than finite differences at $\lambda = 6$, $\Delta t = 0.01$ (verified in `test_weak_form_beats_finite_difference_under_noise`). The weak form of the governing equation for node i is

$$y_i(t) = \lambda\big(u_i(t) - A_{u,i}(t)\big) - A_{I,i}(t) = \sum_{j \in \mathcal{N}(i)} \kappa_{ij}(t)\big(A_{u,j}(t) - A_{u,i}(t)\big). \tag{6}$$

4.4 The identification layer: per-node RLS

Node i maintains a regressor $z_i = [\,A_{u,j} - A_{u,i}\,]_{j \in \mathcal{N}(i)}$ of filtered neighbor differences and fits $y_i \approx a_i \cdot z_i$ by recursive least squares with exponential forgetting ($\alpha = e^{-\lambda\Delta t}$, fast) and a second, much slower consolidation window ($\lambda_s = 0.5$) plus one edge-space smoothing pass, giving the symmetric estimate $w_{ij} = \frac{1}{2}(a_{ij} + a_{ji})$ that feeds morphogenesis. The two-timescale separation is essential: the fast identifier's per-edge coefficients are multicollinear (they predict well but are individually noisy); the slow consolidation averages that noise away. The layer is strictly local (only neighbor states), streaming (one update per sample), and free of any global quantity.

4.5 The morphogenesis layer: angulation

Each node carries a structural vector $v_i \in \mathbb{R}^3$ with spherical parts $r_i = \|v_i\|$ (metabolic magnitude) and orientation (θ_i, ϕ_i). Learning rotates the orientation via nematic consensus toward the coupling-weighted mean direction of neighbors,

$$\hat{v}_i \leftarrow \operatorname{norm}\Big(\hat{v}_i + \eta_e\, g_i\, \frac{\sum_j w_{ij}^{+}\hat{v}_j}{\sum_k |w_{ik}|}\Big), \qquad w_{ij}^{+} = \max(0, w_{ij}), \tag{7}$$

gated by the chemical continuum,

$$g_i = \Theta\Big(\frac{c_i}{\max c}\Big)\left(\frac{c_i/\max c - c_{\text{thr}}}{1 - c_{\text{thr}}}\right)^{p}, \qquad c_{\text{thr}} = 0.6,\ p = 4, \tag{8}$$

plus a concentration-dependent orienting torque toward the drift axis, $\text{rot}_i \leftarrow \text{rot}_i + \eta_t g_i(\hat{w} - \hat{v}_i)$, and homeostatic magnitude regulation $r_i \to r_i + \eta_n(\sqrt{\bar{w}_i} - r_i)$. Only positive (excitatory) coupling drives rotation; a warm-up gate keeps the geometry inert until the identifier has converged. The loop is closed: geometry shapes physics through a_{ij} in κ_{ij}, and physics shapes geometry through the identified w and the chemical field c.

4.6 Documented failure modes

Every stability mechanism exists because a specific failure mode was observed, diagnosed, and patched (Table 6).

4.7 Baselines and metrics

All baselines consume the **same recorded observations**. (i) *Persistence*: $\hat{u}(t+1) = u_{\text{obs}}(t)$. (ii) *Finite-difference RLS*: the identical per-node identifier with raw finite-difference targets, run offline. (iii) *Global batch oracle*: the full per-edge conductivity matrix fit by least squares on the same weak-form-filtered data over the training window, then frozen (best-case global estimator in the correct model class). (iv) *Streaming-global RLS*: the same global per-edge matrix fit with exponential forgetting over the whole trajectory, at two forgetting rates—matched to the local identifier's window (sample-efficiency failure) and tuned slow ($\lambda_g = 0.25$, a ∼200-step window; the best a global streaming model can do). Metrics: physical one-step NMSE vs. the observation-noise floor; law-fit NMSE on the held-out weak-form targets; convergence latency (first time the rolling law-fit NMSE stays below 0.2); alignment statistics; read-back correlations; the axial Dirichlet routing probe with a random-angles ablation; and runtime memory accounting (measured object sizes).

4.8 Hardware mapping (design rules)

1. **IIR weak-form filter** → an RC ladder (one-pole exponential window) per node; the weak derivative $y = \lambda(f - A)$ is a local subtraction—no differentiators.
2. **Per-node RLS** → a small analog adaptive filter per node (forgetting = RC time constant; ridge = leak). All state is $O(d^2)$ per node, constant in trajectory.
3. **Electrical continuum** → a resistive network; the implicit update is the physical settling of the network itself (capacitance)—the material *is* the solver.

Table 6: Failure modes encountered and their patches.

#	Failure (observed)	Diagnosis	Patch
F1	Bilinear per-node RLS collapses to $v = 0$	The regressor contains neighbor parameters; exact LS attributes the full target to v_i, mapping $v \to v/2$ per update	Gradient-form rotation (Hebbian angulation) instead of exact bilinear LS
F2	Early geometry destruction (alignment $0.6 \to 0.15$)	Rotating on noise-dominated early weights is a random walk	Warm-up gate: morphogenesis engages after identification converges
F3	Negative-w anti-aligned checkerboard	Identification noise produces negative couplings; rotating along them anti-aligns everything	$w^+ = \max(0, w)$ in the rotation; homeostatic norm pinning
F4	Decimated slow stats explode to $w \approx -24$	Decimated EMA used the per-step forgetting per 5-step update, stretching the window 5× and freezing stale early data	Stride-compensated forgetting α^{stride}
F5	Explicit-Euler divergence at high conductivity	$\kappa\Delta t$ deg approaches the explicit stability limit as alignment and chemistry grow	Implicit port-Hamiltonian solve (unconditionally stable)
F6	Directional beam-gate wiring forms broken chains	Softmax single-winner aiming makes pairwise alignment only indirect; chains break, killing routing	Revert to alignment coupling a_{ij}; nematic consensus rotation (a contraction)
F7	Consensus rotation globalizes the shape	The consensus propagates across the whole grid; mean alignment $\to 1.00$ and $\text{corr}(a_{ij}, \kappa) \to 0.1$—a uniform κ boost changes no potential field	Chemical gate g_i: morphogenesis is zero below a concentration threshold; the shape is a focused corridor
F8	Routing probe collapses to $u = 0$	Dirichlet conditioning zeroed the pinned columns without moving the known potentials to the RHS, severing source/sink from every equation—all "routing" was a boundary-layer artifact	Move pinned potentials to the RHS before zeroing rows/columns (`route_potential`)

4. **Chemical continuum** → a diffusive/fluidic layer (or RC diffusion lines with drift) whose local concentration modulates edge conductance—memristive elements with concentration-controlled conductance.

5. **Angulation** → each connection's direction is a pair of analog phase/amplitude values; learning rotates the phase. Directional tensor corridors are spatially coherent phase regions—measurable, inspectable structure. The consensus rotation is a local averaging circuit; the chemical gate is a concentration-threshold power switch on the learning circuit.

6. **Warm-up and homeostatic dampening** → power-gated or time-constant-gated learning and per-element leak.

5 Data and code availability

All experiments are deterministic (seeds via `numpy.random.default_rng`) and reproducible with:

```
pip install numpy matplotlib
python run_experiments.py --task sweep    --seeds 15 --T 4000 \
       --out results.json
```

```
python run_experiments.py --task ablation --seeds 8  --T 4000 \
       --out ablation.json
python run_experiments.py --task scale    --seeds 5  --T 4000 \
       --sizes 7x7,11x11,15x15 --out scaling.json
python make_figures.py
python test_biomaterial_net.py
```

The three result files committed alongside the code are the exact outputs of these commands; the manuscript's numbers are read from them. Configuration: `default_cfg()` in `biomaterial_net.py`, documented inline.

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